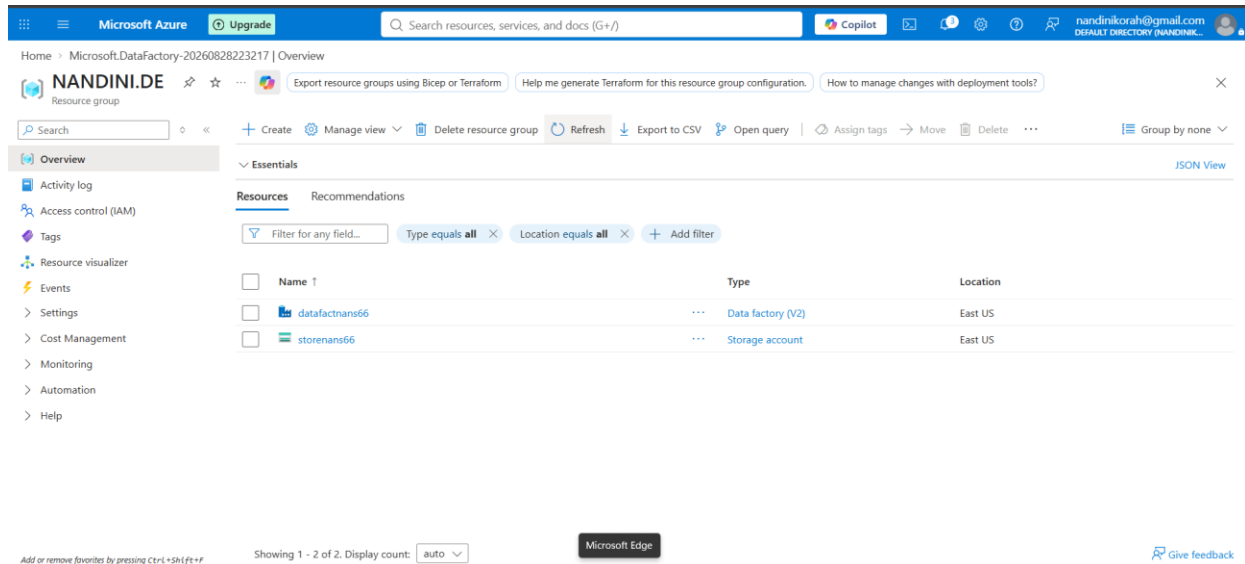


# SERP Scrapping for Live Google News

Using SearchApi.io → Azure Data Factory → ADLS Gen2

## 1. Overview

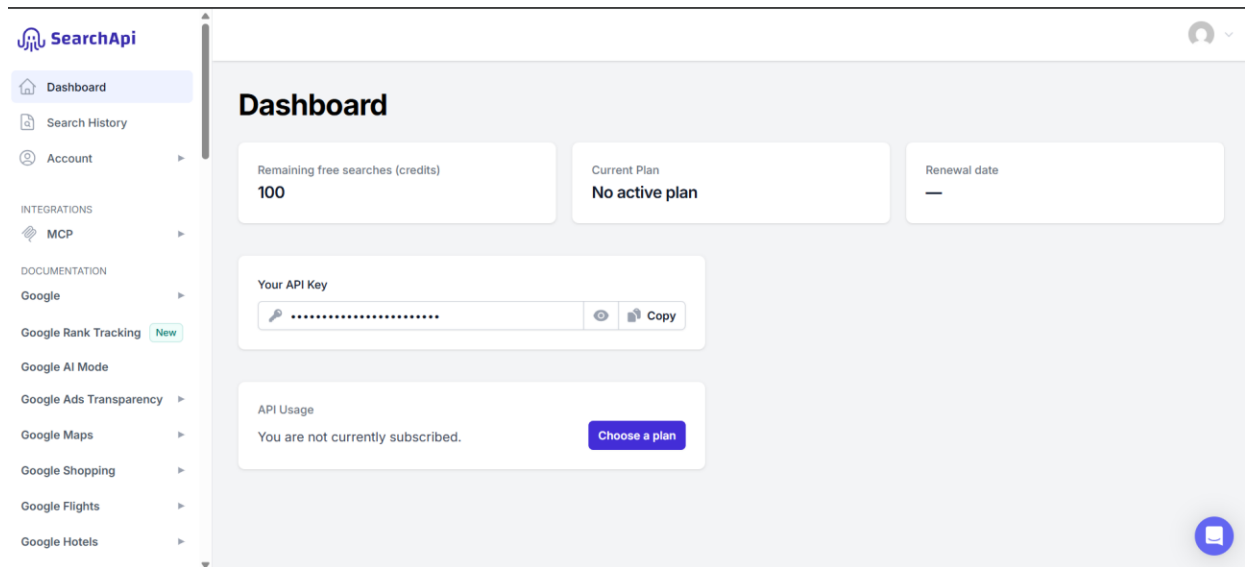
This project fetches live Google News results via SearchApi.io (SERP API) and stores them as JSON in Azure Data Lake Storage, orchestrated using an Azure Data Factory pipeline.



## 2. Get the SearchApi.io API Key

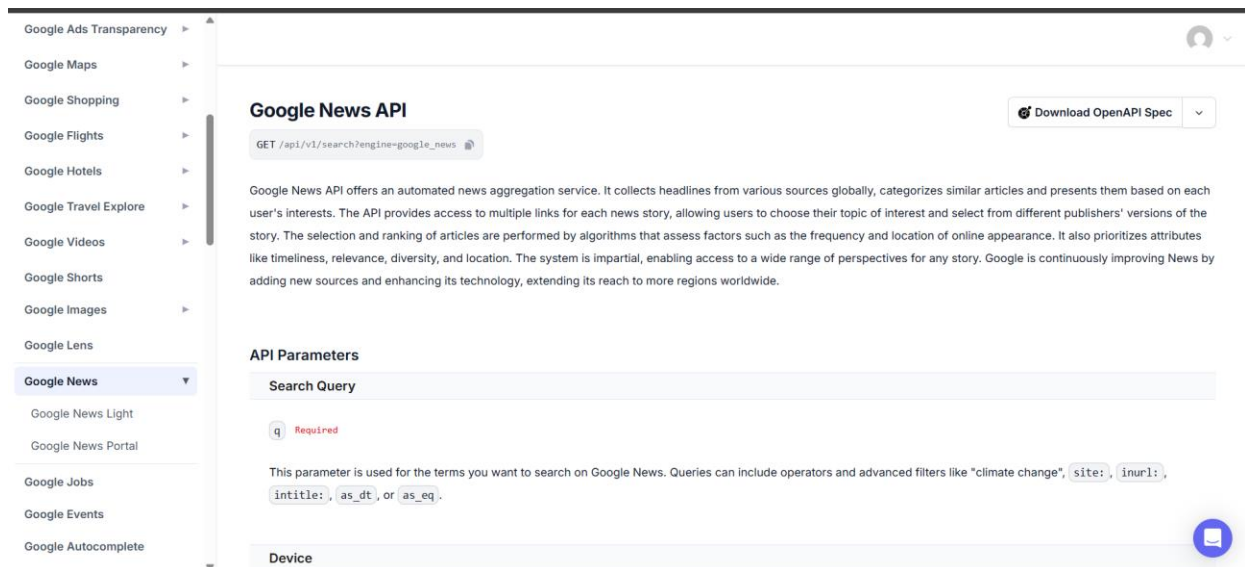
### Step 1: Sign up / log in to SearchApi.io

Copy the API key from the Dashboard — needed for the REST linked service.



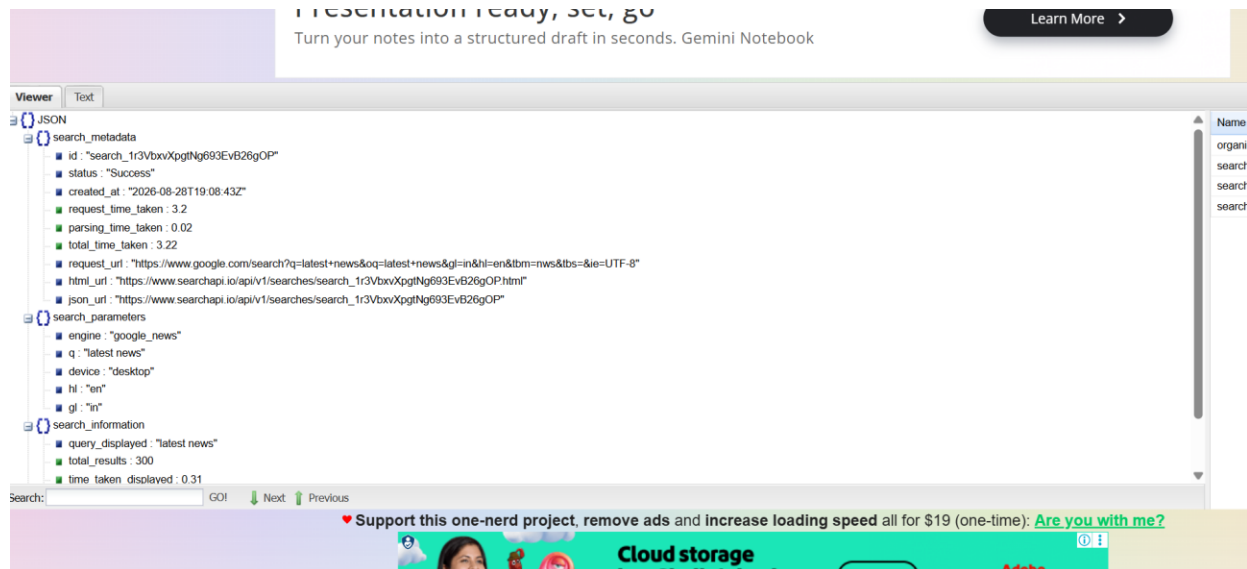
## Step 2: Review Google News API docs

Check required parameters: engine=google\_news, q (search query), hl, gl.



## Step 3: Sample JSON response

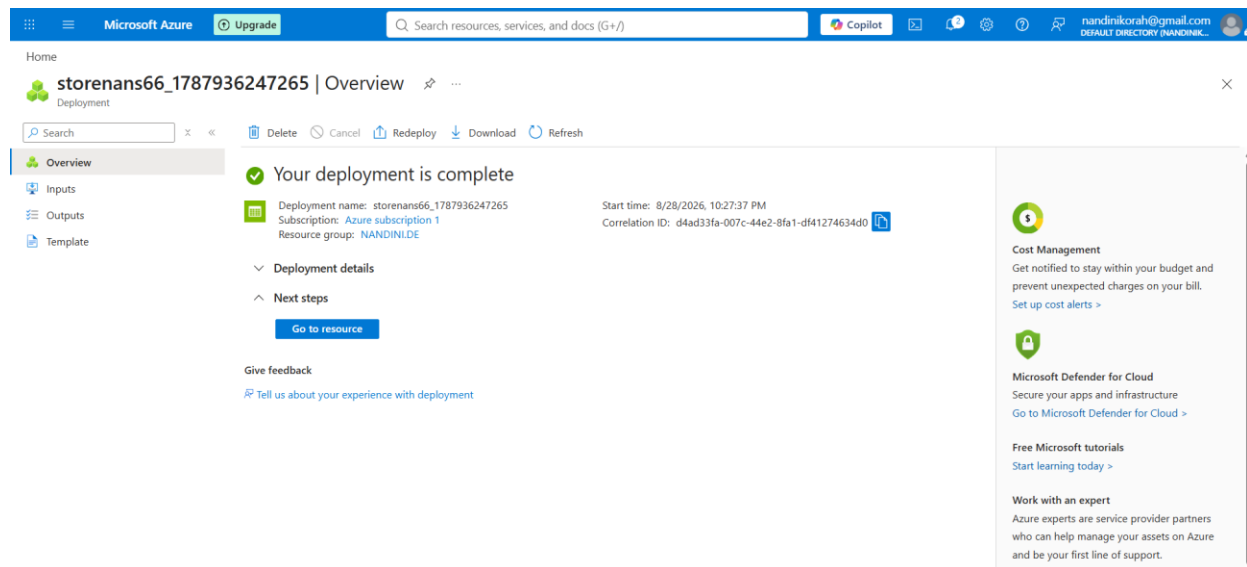
Preview the raw response structure returned by the API.



### 3. Create Azure Storage Account (ADLS Gen2)

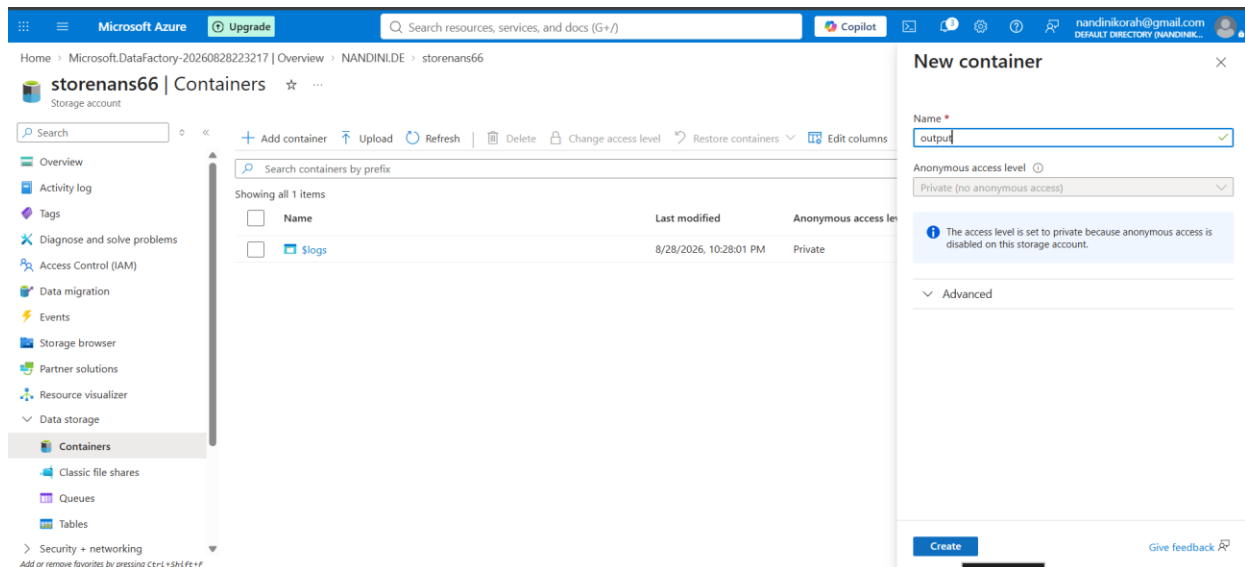
#### Step 4: Deploy storage account

Deployment completes successfully alongside the Data Factory.



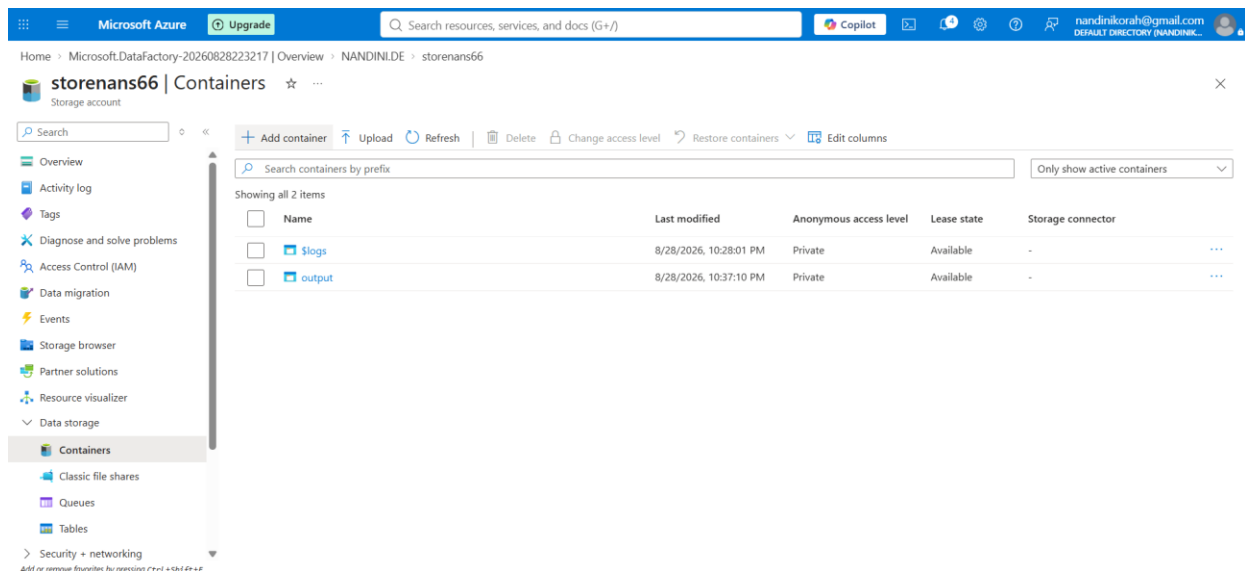
#### Step 5: Create 'output' container

Add a new blob container named 'output' to store the JSON results.



## Step 6: Verify container created

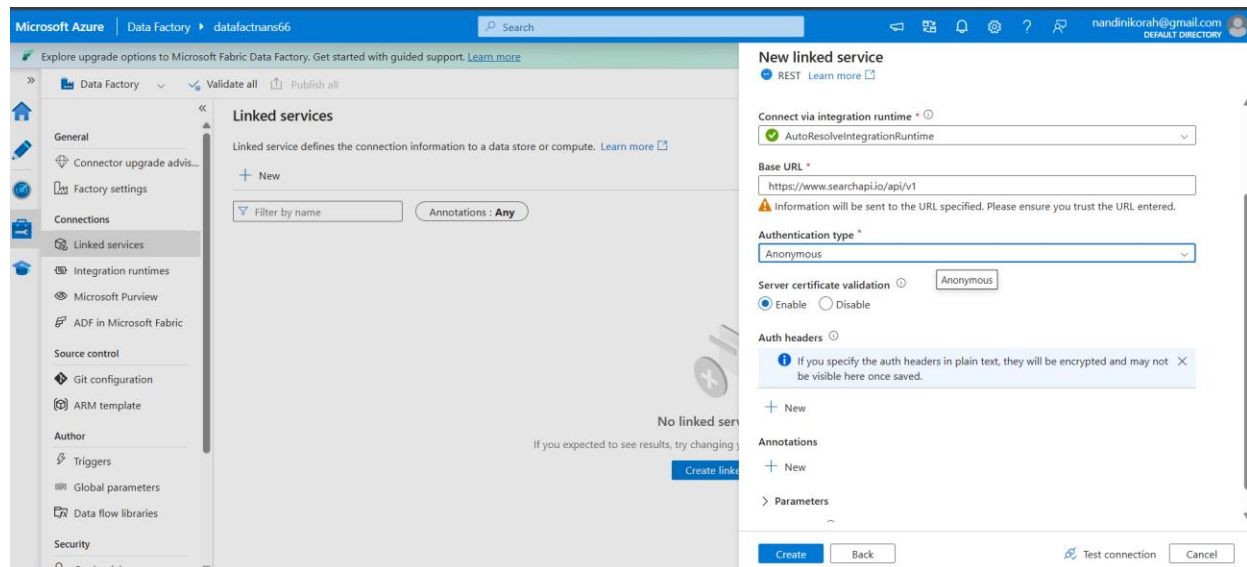
The 'output' container now appears in the storage account.



## 4. Configure Linked Services in ADF

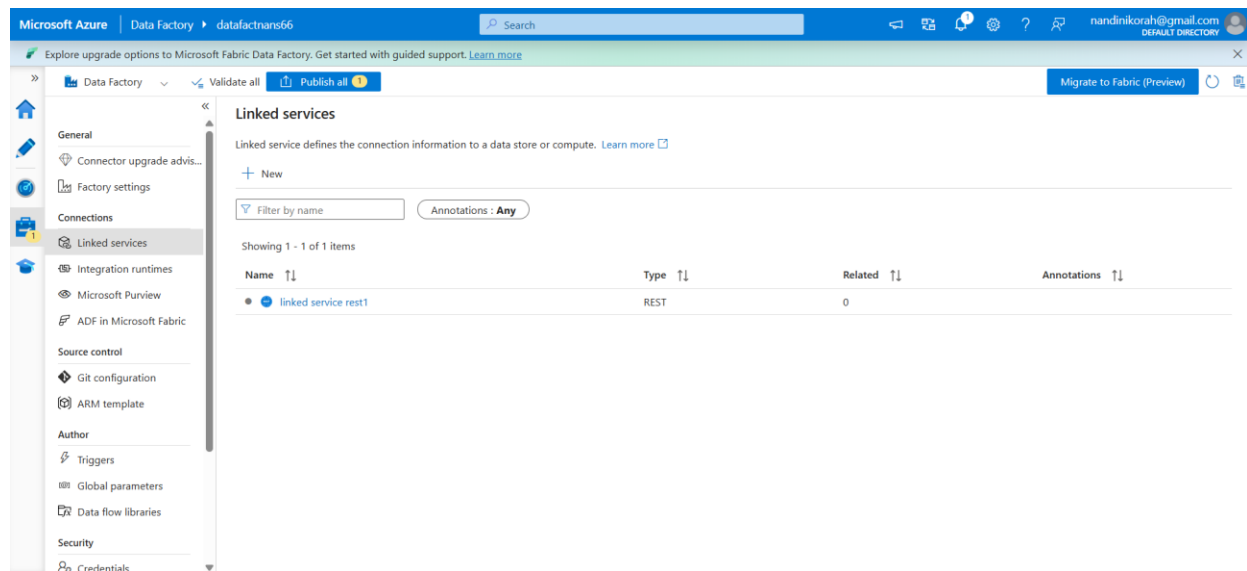
### Step 7: Create REST linked service

Base URL: <https://www.searchapi.io/api/v1>, Authentication: Anonymous (API key passed via query param).



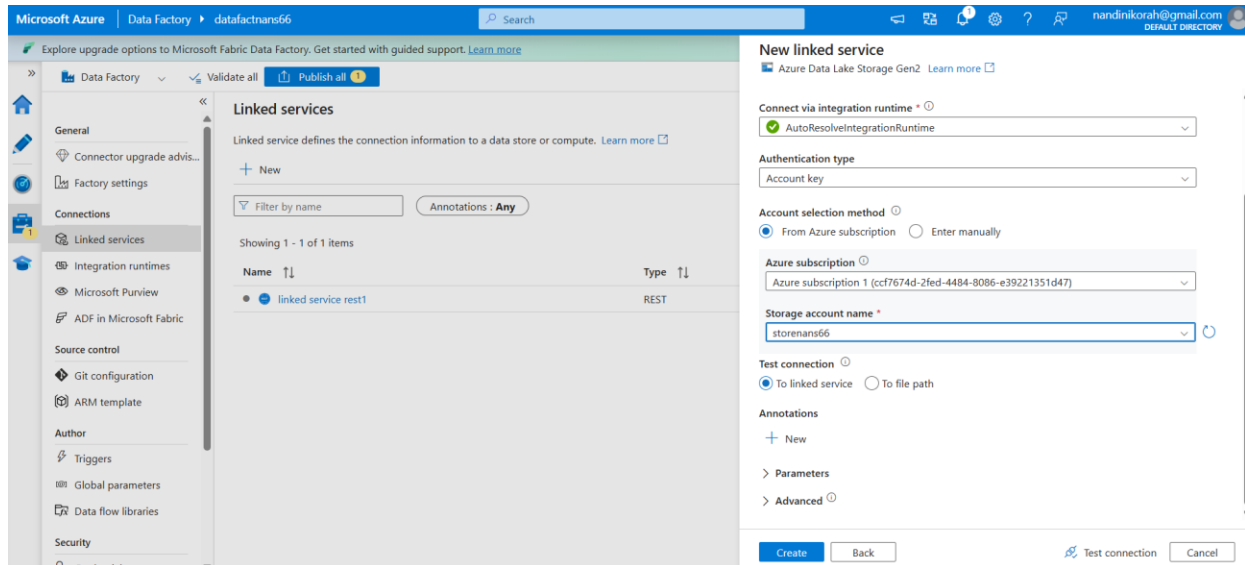
## Step 8: Linked service created

'linked service rest1' appears under Linked Services.



## Step 9: Create ADLS Gen2 linked service

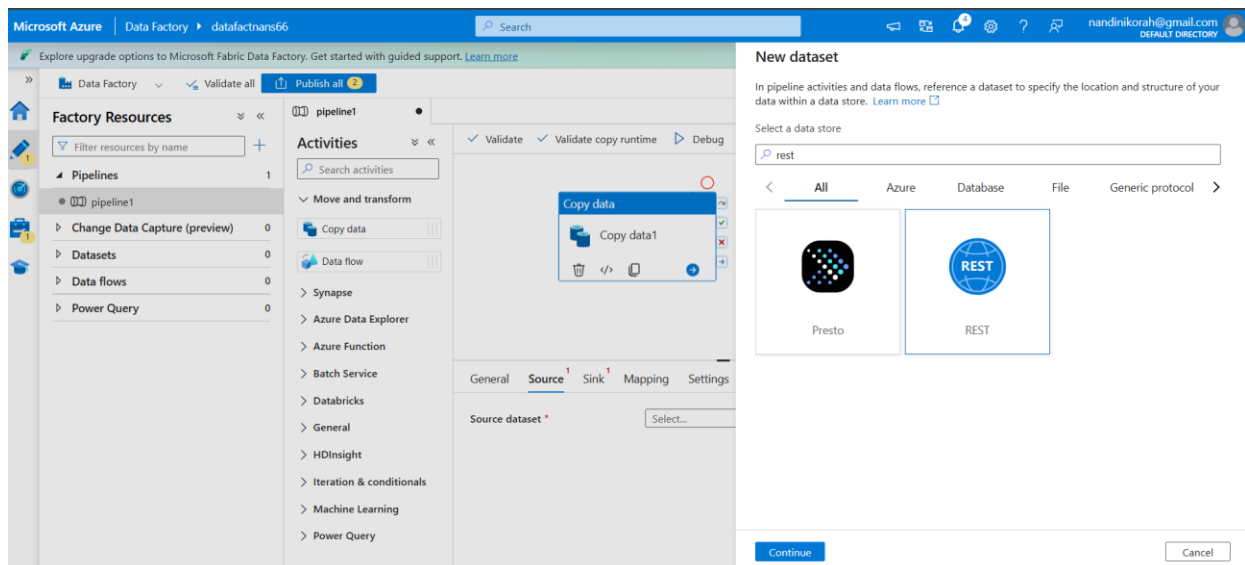
Connect using Account key, selecting the storage account 'storenans66'.



## 5. Create the REST Source Dataset

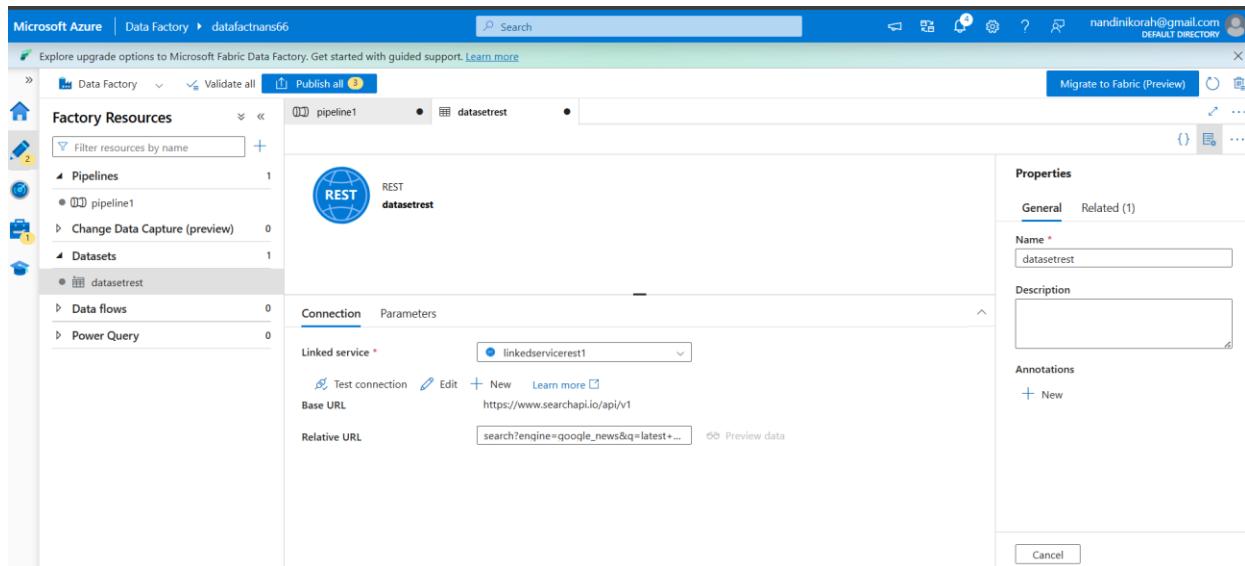
### Step 10: New dataset → REST

Select REST as the data store type for the source dataset.



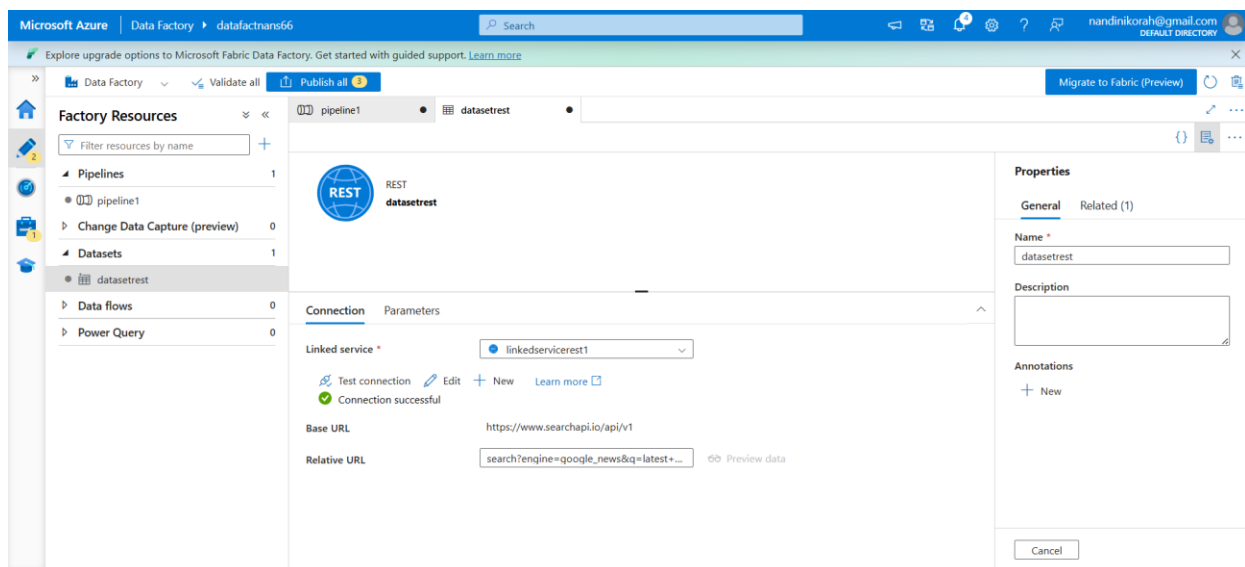
## Step 11: Set Relative URL

search?engine=google\_news&q=latest+news&gl=in&hl=en, using linked service 'linkedservicorest1'.



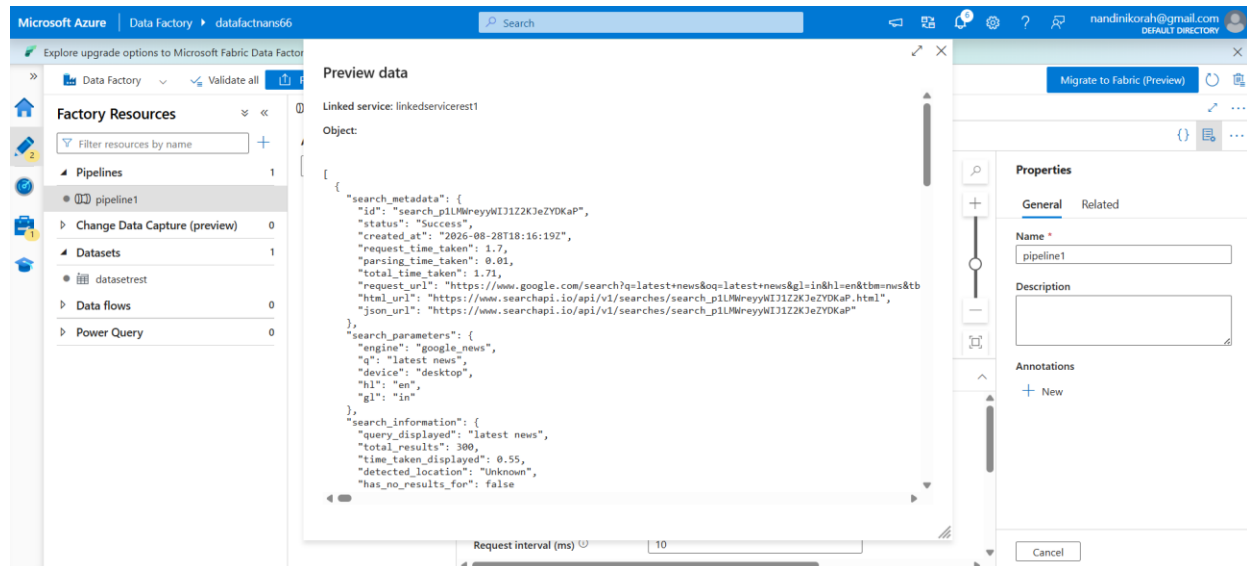
## Step 12: Test connection

Connection to the REST API succeeds.



## Step 13: Preview data

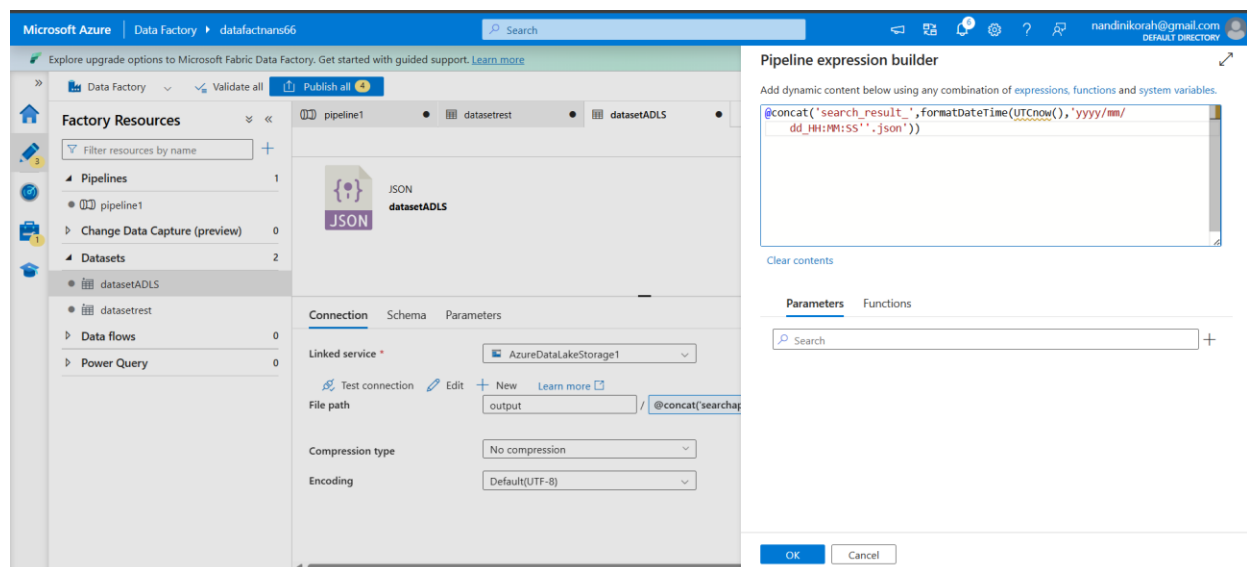
Confirms live JSON news data is returned from the API.



## 6. Create the ADLS JSON Sink Dataset

### Step 14: Build dynamic file path

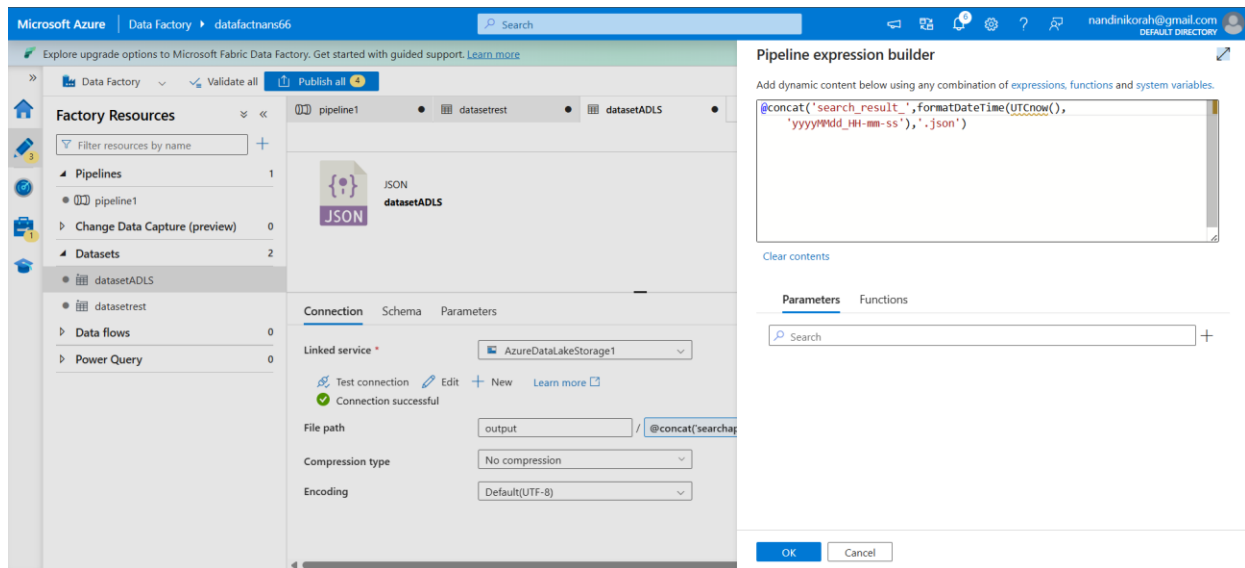
Use pipeline expression builder for folder path: `concat('searchapi/google/latest+news/', ...)`.





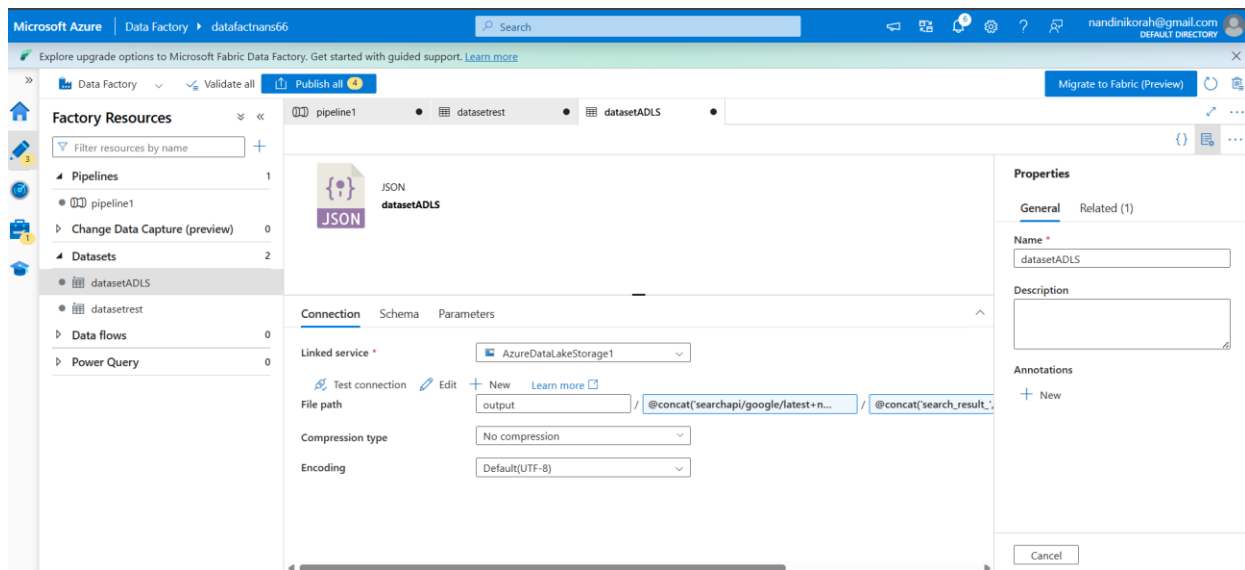
## Step 15: Correct the filename expression

`concat('search_result_', formatDateTime(utcnow(),'yyyyMMdd_HH-mm-ss'), '.json')` for a unique file name per run.



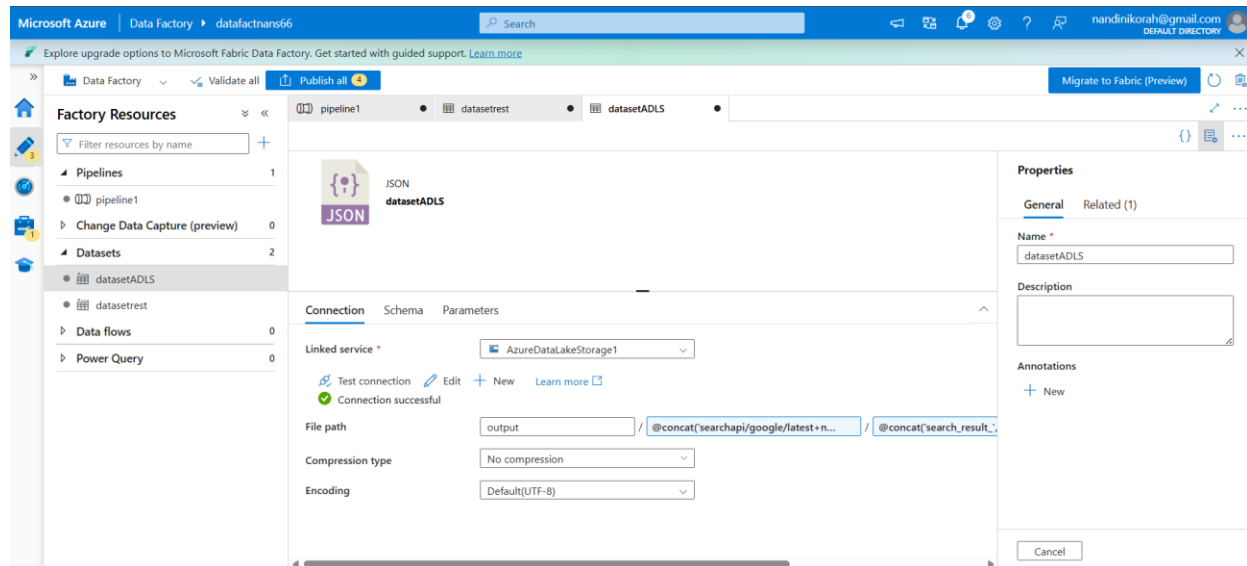
## Step 16: Final dataset connection

datasetADLS points to container 'output' with the dynamic folder/file path.



## Step 17: Dynamic paths applied

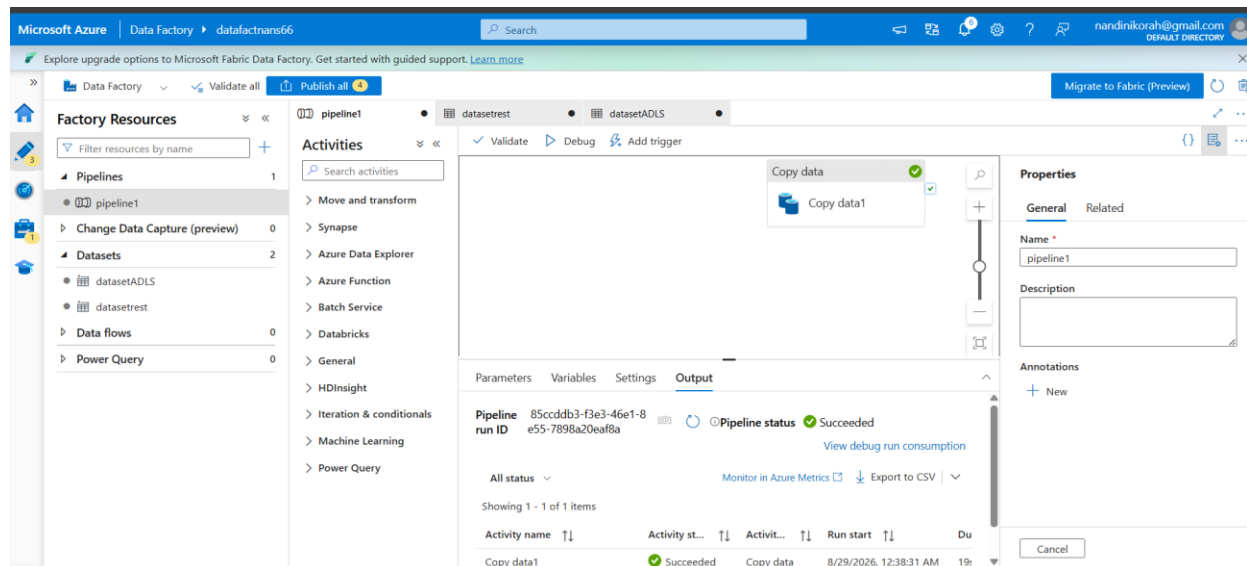
Both folder and file name expressions are set on the ADLS dataset.



## 7. Build and Run the Pipeline

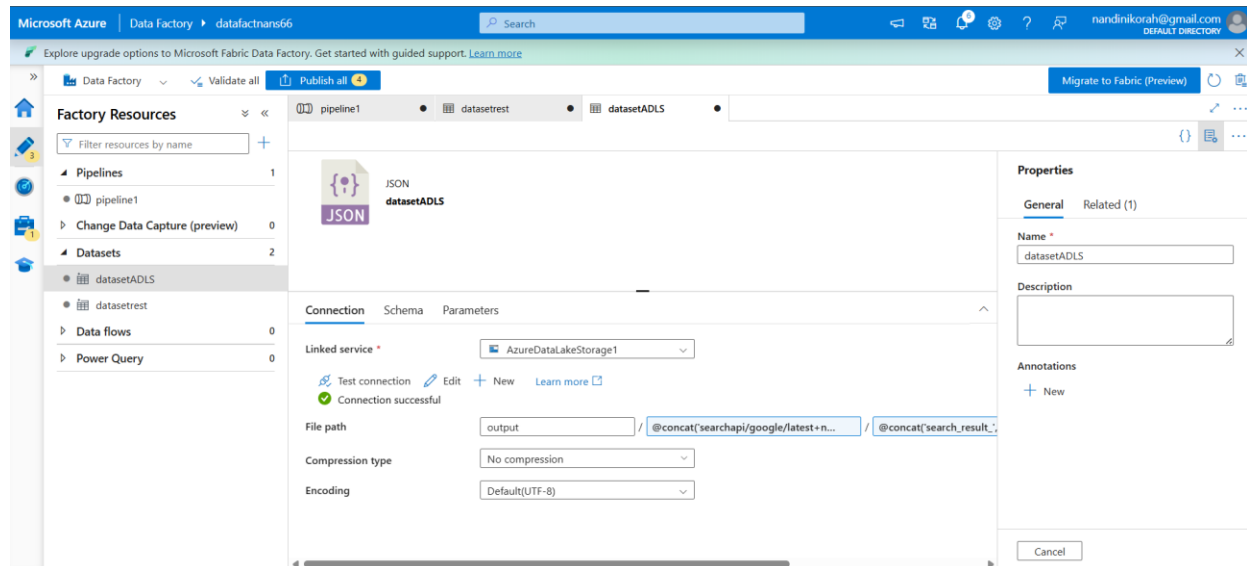
### Step 18: Copy Data activity

Source: datasetrest (REST), Sink: datasetADLS (JSON) inside 'pipeline1'.



## Step 19: Verify dataset in pipeline

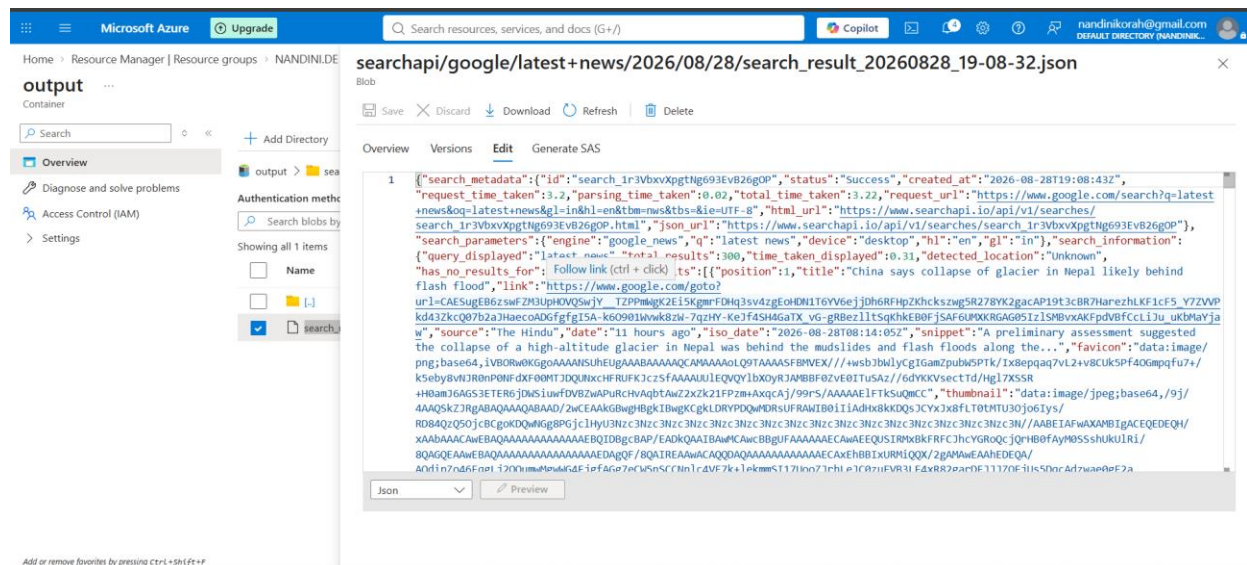
datasetrest connection is validated as successful in the pipeline context.



## 8. Verify Output in Blob Storage

### Step 20: Check output JSON file

The pipeline run creates a timestamped JSON file inside output/searchapi/google/latest+news/... containing live news results.



## Result

Live Google News data is fetched from SearchApi.io via a REST linked service and stored as JSON in Azure Data Lake Storage, fully automated through an Azure Data Factory pipeline.